

Chapter 46: The Emerging Multidisciplinary Trends in Dermatoglyphics: The Science of Palmar and Plantar Configurations

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Abstract

Dermatoglyphics: A Transdisciplinary scientific approach of anthropology, dermatology, genetics, forensics, computer science, and medicine relies on the palmar and plantar configurations confined to the finger extremities, palms, and soles. Transdisciplinary methodologies in dermatoglyphic studies endorsed and correlated atypical dermatoglyphics with derailed genomic machinery, unnatural intrauterine ecosystems, and embryonic progression. The fingerprint patterns, palmar 'atd' angle, finger ridge count, total finger ridge count, absolute finger ridge count, and position of axial triradius are the fundamental qualitative and quantifiable parameters examined and enumerated in dermatoglyphic studies. Erroneous palmar, plantar, hypothenar, and thenar patterns in congenital anomalies have been recorded. Diagnostic and prognostic dependability and reliability of irregular dermatoglyphics and asymmetric volar configurations are universally verified. Instigated as an invasive and simple tool of personal identification, dermatoglyphics is currently evolving in transdisciplinary science areas of forensics and criminal investigation. A collaborative approach in dermatoglyphic research enabled the advancement of dermatoglyphics into applied dermatoglyphics. Exploration of the uniqueness of racial and tribal palmar configurations paved the way for ethnic dermatoglyphics.

Keywords: Dermatoglyphics, Fingerprint, Palmar, Plantar, Transdisciplinary.

1. Introduction

Dermatoglyphics: the science of exploration of friction ridges and configurations confined utterly to the finger balls, palmar and plantar surfaces. Epidermal ridges and furrows characteristically reported in primates as friction ridges. Sir Francis Galton (1892) paved foundation of dermatoglyphics by asserting the inalterability and unchangeability of the palmar and plantar ridges, collectively called dermatoglyphs, stands polygenic, unique and parallelly aligned. Dermatoglyphics the “science of fingerprints” prehistorically emerged as non-invasive and economical tool for personal identification. During 1960s Cummins and Midlo pioneered the emergence of dermatoglyphics as diagnostic and prognostic markers for various congenital anomalies. Transdisciplinary approaches transformed the dermatoglyphics mere science of fingerprints into ‘forensics’, ‘biometrics science’ and ‘science of health’. Anthropological approach of dermatoglyphics unlocked ethnic dermatoglyphics focused on the comparative analysis of palmar configurations of races and ethnic groups. Science of biometrics is fundamentally instigated from dermatoglyphic measurements. Anomalous and atypical palmar patterns have been described in several inborn ailments with substantial empirical evidence (Schanmann & Alter, 1976). Emergence of dermatoglyphics as reliable marker for the prognostication of congenital anomalies considered to be an outcome of transdisciplinary approaches and comparative evaluation palmar symmetry (Cummins & Midlo, 1943). Dermatoglyphs originate ectodermally during the 13th week of gestation from the embryonic rudimentary volar pads (Babler, 1991). Antenatal differentiation and establishment of the friction ridges on the finger extremities, palms get through by 16th week of gestation (Kucken et al.,

2005). Embryonic advancement of the volar pads into the epidermal ridges are restrained by genomic machinery. Intrauterine ecosystems are critically influential in instigation and differentiation of palmar and plantar volar configurations. Irregular and atypical dermatoglyphic patterns reflect erroneous and inaccurate gene machinery, intrauterine disarray and defective antenatal embryonic progression. Transdisciplinary approaches of different academic domains enhanced the outlook and potential of dermatoglyphics. Integration of medicine and dermatoglyphics has opened possibility of noninvasive and economic methodology of prognosis.

2. Dermatoglyphic Research an Outline:

Dermatoglyphic studies exclusively depend on dermal ridges located on the finger balls, configural parameters on the palmar digital creases, thenar and hypothenar, interdigital area of palm. Fingerprint patterns engraved on the finger extremities are the major qualitative parameter in dermatoglyphic research. The 'atd' angle, finger ridge count, total finger ridge count, absolute finger ridge count, position of axial triradius are the fundamental parameters examined and enumerated quantitatively in dermatoglyphic studies. Fingerprint patterns were initially suggested by Sir Francis Galton (1982) based on the opening and flow of ridges, in his famous book *Fingerprints*, 1980, as the arch, the loop and the whorl. Edward Henry revised the finger patterning by incorporating the concept of 'core' and 'delta' in finger patterns, thus sub-grouped Galton system (Yager&Amin, 2004). The 'atd' angle is yet another quantitative measure of dermatoglyphics; hypothetical geometry formed in the palm (Brunson et al., 2015). Finger ridge counts, total finger ridge count, absolute finger ridge count and palmar ridge count are being assessed in dermatoglyphic studies. Location and displacement of axial triradii and its distal displacements quantitatively measured.

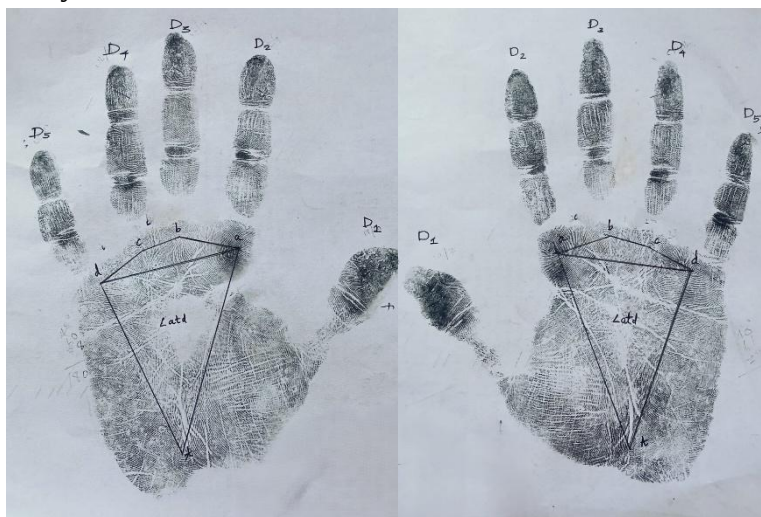


Figure 1: Palmprints showing digits (D1 to D5) and angle 'atd'

Researchers documented atypical fingerprint patterns and distorted dermatoglyphics in various congenital anomalies. The relationship between abnormal dermatoglyphics and genomic inaccuracy, extra and intrauterine environment and embryonic progressions were documented. Penrose (1968) elucidated and significantly correlated the altered palmar dermatoglyphic configurations with congenital chromosomal aberrations, mainly the polyploid conditions. Abnormally deviated in palmar, plantar, thenar, hypothenar and interdigital dermatoglyphs have been reported Down's Syndrome, Klinefelter's Syndrome, Edward's Syndrome, Turner Syndrome, Cri du chat Syndrome, Patau Syndrome, and Noonan

Syndrome (David,1963; Purvis, 1972; Shiono&Azumi, 1982; Lakshmi Prabha&Thenmozhi, 2014; Fujita, 1968; Plato et al.,1973; Hassanzadeh et al.,2007; Cummins, 1939; Bryant et al.,1970). Abnormal dermatoglyphs were documented in congenital Maxo-facial misconfigurations (Saxena et al., 2013) cardiac anomalies (Tay et al.,1985), renal absurdities (Wijerathne et al.,2016) and diabetes (Shrestha et al.,2023). Uchida (1967) endorsed the endorsed dermatoglyphics in screening and prognostication of the diseases, undoubtedly recommended the reliability and trustworthiness of dermatoglyphics as a marker.

3. Multidisciplinary Approaches in Dermatoglyphics:

3.1 Medical Dermatoglyphics

Transdisciplinary medical studies have universally supported clinical significance of dermatoglyphics and widely approached dermatoglyphics as method of prognostication. Renowned dermatoglyphist Penrose suggested the utilization of friction ridges on the finger balls and palms for diagnosis and genetic studies. From the light of thorough dermatoglyphic investigations on human polyploidy aberrations, Penrose (1968) Influence of genomic factors and intrauterine environment in formation dermatoglyphics were of undisputedly revealed, proposed dermatoglyphics as biological marker for various congenital anomalies. Integrative modern medicine has explored the role of papillary ridges on palms and soles as non-invasive and economical screening tools in identification of early risk factors (Tewari et al., 2002). Medical genetics referred dermatoglyphics as genetic markers contemplate the intrauterine ecosystem and embryonic progression. Hereditary and genomic potentiality of dermatoglyphics were assessed in the elucidations of Borgaonkar et al., (1973) and David(1973). Aforesaid studies documented dermatoglyphics as hereditary and genomic marker. Interdisciplinary studies conducted in behavioural psychology and dermatoglyphics documented significant relationships. Mellor (1992) described increasing asymmetries in fingerprint patterns and finger ridge counts among the schizophrenia patients. Extensive research has been conducted exclusively among the schizophrenia patients and observed atypical dermatoglyphs (Latif,2017; Markow & Wandler,1986).

3.2 Forensic Dermatoglyphics

Forensic dermatoglyphics; transdisciplinary subsidiary relies on the dermatoglyphic configurations for criminal investigation and personal identification. Invariancy, permanency and uniqueness of fingerprint patterns and palmar elevations are being exploited to find criminals, suspects and victims. The fingerprints are the most valuable, trusted and possible form of physical evidence that forensic investigators rely on. O'Neill (1940) classified the fingerprints forms, molded impressions, visible impressions, and latent impressions where the latent types are widely used in the forensic studies. Transdisciplinary methodologies have been exploring to develop and prepare the invisible latent fingerprints moreover to enhance efficacy and preciseness. Sahu & Jain (2025) reported application of nanoparticles in fingerprint identification, claimed that the aforesaid enhance the sensitivity and selectivity in criminal site investigation. Sametband et al., (2007), Singh et al., (2024), Bhati & Tripathy (2020) and many other researchers have suggested the multivariant application of nanoparticles in fingerprint elucidation to enhance the reflectivity there by accuracy in fingerprint recognition.

3.3 Anthropometric Dermatoglyphics

Transdisciplinary integration of anthropology and dermatoglyphics made significant revolutions in the pursuit of anthropology. Comparative dermatoglyphics made ethnic variation and ethnic affinity studies more scientific and precise. Dermatoglyphs features of inalterability had been brilliantly exploited by the researchers to establish relationships between tribes and micro-ethnic groups. Multivariate assessments in the anthropological dermatoglyphic dimensions enabled them to discover the evolutionary relationship between humans and non-human primates, apes (Dorjee et al., 2014). Dermatoglyphic distinctiveness among the different populations and tribes were documented by Thoma (1969) and Roberts (1972). Thoma (1969) has documented the significant differences in finger ridge count, a quantitative measure of the palmar dermatoglyphics among a Hungarian population. Das B.M an Indian researcher performed numerous dermatoglyphic-anthropology collaborative studies among various tribes, populations, caste groups (Dorjee et al., 2014).

3.4 Dermatoglyphic Topology

Dermatoglyphic topology is the interdisciplinary integration of mathematics and dermatoglyphics where the designs and computations are employed to understand and interpret the dermal patterns. Dermatoglyphic studies in topological dimensions enable the researchers to interpret the patterns and configurations more precisely than mere quantitatively. Execution mathematical algorithms to assess the qualitative as well as quantitative parameters of dermatoglyphics. Qualitative configuration like fingerprint pattern elucidating based on the number of loops and triradii in dermatoglyphic topology. Penrose (1965) proposed a comprehensive proposal dermatoglyphic topology, suggested to approach dermatoglyphics with mathematical formulae. Dermatoglyphic topology made qualitative assessments more precise. Penrose (1965) established relationship between the number of loops in fingerprints and number of triadius.

3.5 Biometric dermatoglyphic

Biometric dermatoglyphic refers to an integrative domain of biometric identification is momentarily progressing in the current scenario, depending on the fingerprints and palmar configurations. Modern biometrics aided with dermatoglyphics is critically useful in accurate personal identification, authentication and security. Biometric identification using the fingerprints have digitized and new finger pattern recognition systems were updated. Various stages of fingerprint biometrics such as fingerprint acquisition, image processing, locating distinctive characteristics, template creation and template matching has been updated (Lourde & Khosla, 2010). Fingerprint acquisition has been modernized with deployment of optical scanner and capacitive scanner, thermal scanner and ultrasound scanner. Procedures of fingerprint matching have been revolutionized with the updated minutiae-based automated fingerprint identification systems, providing more accuracy and high interpretability (Maiti et al., 2025). Modern dermatoglyphics has pore based fingerprint recognition, process high accuracy and (Xu et al., 2010).

3.6 Psychometric dermatoglyphics

Psychometric dermatoglyphics refers to an integrative domain of psychometrics and dermatoglyphics, analysis the dermatoglyphs to assess the degree of intelligence, personality, skills, emotions and behaviour. Relationship between the palmar dermatoglyphic configurations and cognitive aptitudes has been proven in many studies. Various psychometrics tests were formulated to assess cognitive aptitudes

based on fingerprints and palmar configurations. Tharay et al., (2020) performed an extensive investigation to assess the relationship between the intelligence quotient (IQ) and dermatoglyphic configurations on finger balls and palms. Aforesaid study of Tharay et al., (2020) established a significant connection between the fingerprint pattern and IQ. The studies of Najafi (2009), Cesarik et al., (1996), and Kiran et al., (2010) have documented the relationship between the fingerprint patterns and IQ. Researchers proposed dermatoglyphic assessments for diagnosing the lowering IQ or mental retardation (Kiran et al., 2010). Dermatoglyphic and Multiple Intelligence Test (DMIT) is controversial psychometric test, not scientifically proven yet, assessing the cognitive capabilities, reportedly based on the Howard Gardner's theory (Abhimanyu, 2016).

4. Conclusion

Transdisciplinary integration deepens the academic depth moreover multifaceted approaches elaborate the potential and possibilities. Dermatoglyphics was instigated as a science of personal identification based on fingerprint patterns however transdisciplinary approaches transformed the pursuit of dermatoglyphics into broad and extensive. Incorporation of medical sciences in dermatoglyphics transmuted dermatoglyphics into marker and tool of prognostication. Amalgamation of different academic areas such as anthropological science, genetics, biometrics, and topology transfigured classical dermatoglyphics into transdisciplinary dermatoglyphics, much capable of doing wonders for humankind, from personal identification to prognostication of abnormalities

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